Coursework 1

Mathematical Modelling in Biology, Medicine and Public Health

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1. (a) **SIR model.** At a disease-free equilibrium, $S \approx N$, so the infected equation becomes $\dot{I} \approx \beta_2 I - (g_2 + d)I$. At this equilibrium, the per-capita infection rate is approximately β_2; the rate at which one leaves the infectious class is approximately $g_2 + d$, thus the mean infectious period is $\frac{1}{g_2+d}$. Hence,

$$\boxed{R_0^{\mathrm{SIR}} = \frac{\beta_2}{g_2 + d}}.$$

SEIR model. Newly infected individuals must first enter E and later progress to I. Since exposed individuals are not infectious, we must account for the probability that an exposed person survives the latent stage and reaches an infectious state:

- Rate of leaving E: $a + d$.
- Rate of progressing from E to I: a.

So, the probability that an exposed individual reaches an infectious state is $\frac{a}{a+d}$. Once in I, the transmission rate is β_1 and the mean infectious period is $\frac{1}{g_1+d}$. Therefore,

$$\boxed{R_0^{\mathrm{SEIR}} = \frac{a\beta_1}{(a+d)(g_1+d)}}.$$

(b) **SIR model.** To determine the endemic equilibrium, we set $\dot{S} = \dot{I} = \dot{R} = 0$. From the infected equation, $0 = \beta_2 \frac{S^* I^*}{N} - (g_2 + d)I^*$. Since $I^* > 0$ at the endemic equilibrium, dividing by I^* gives $\beta_2 \frac{S^*}{N} = g_2 + d$. Thus, $S^* = \frac{N(g_2+d)}{\beta_2}$. Next, from $\dot{S} = 0$, $0 = dN - \beta_2 \frac{S^* I^*}{N} - dS^*$. Using $\beta_2 S^*/N = g_2 + d$, we obtain $0 = dN - (g_2 + d)I^* - dS^*$, hence $(g_2 + d)I^* = d(N - S^*)$ and

$$I^* = \frac{d(N - S^*)}{g_2 + d}.$$

Finally, from $\dot{R} = 0$, $0 = g_2 I^* - dR^*$, giving $R^* = \frac{g_2}{d} I^*$. Therefore, the endemic equilibrium is

$$\boxed{(S^*, I^*, R^*) = \left(\frac{N(g_2 + d)}{\beta_2}, \frac{d(N - S^*)}{g_2 + d}, \frac{g_2}{d} I^* \right)}.$$

Note that this equilibrium exists iff $R_0 > 1 \iff S^* < N$. To compute the Jacobian, define $f_1 = dN - \beta_2 \frac{SI}{N} - dS$, $f_2 = \beta_2 \frac{SI}{N} - (g_2 + d)I$ and $f_3 = g_2 I - dR$. Thus, the Jacobian matrix is

$$J_{\text{SIR}}(S, I, R) = \begin{pmatrix} -\beta_2 I/N - d & -\beta_2 S/N & 0 \\ \beta_2 I/N & \beta_2 S/N - (g_2 + d) & 0 \\ 0 & g_2 & -d \end{pmatrix}.$$

Evaluating at the endemic equilibrium, where $\beta_2 S^*/N = g_2 + d$, gives

$$\boxed{J^*_{\text{SIR}} = \begin{pmatrix} -\beta_2 I^*/N - d & -(g_2 + d) & 0 \\ \beta_2 I^*/N & 0 & 0 \\ 0 & g_2 & -d \end{pmatrix}}.$$

SEIR model. We set $\dot{S} = \dot{E} = \dot{I} = \dot{R} = 0$. From the infected equation, $0 = aE^* - (g_1 + d)I^*$, so

$$E^* = \frac{g_1 + d}{a} I^*.$$

From the exposed equation, $0 = \beta_1 \frac{S^* I^*}{N} - (a + d)E^*$. Substituting E^* gives $\beta_1 \frac{S^* I^*}{N} = (a + d) \frac{g_1 + d}{a} I^*$. Since $I^* > 0$, dividing by I^* gives $\beta_1 \frac{S^*}{N} = \frac{(a+d)(g_1+d)}{a}$. Thus,

$$S^* = \frac{N(a + d)(g_1 + d)}{a\beta_1}.$$

From $\dot{S} = 0$, $0 = dN - \beta_1 \frac{S^* I^*}{N} - dS^*$. Using the relation above, $0 = dN - \frac{(a+d)(g_1+d)}{a} I^* - dS^*$. Hence,

$$I^* = \frac{ad(N - S^*)}{(a + d)(g_1 + d)}.$$

Finally, $E^* = \frac{g_1+d}{a} I^*$ and $R^* = \frac{g_1}{d} I^*$. Thus, the endemic equilibrium is

$$\boxed{(S^*, E^*, I^*, R^*) = \left(\frac{N(a + d)(g_1 + d)}{a\beta_1}, \frac{g_1 + d}{a} I^*, \frac{ad(N - S^*)}{(a + d)(g_1 + d)}, \frac{g_1}{d} I^* \right)}.$$

Again, we assume the existence of $R_0 > 1$. Define $f_1 = dN - \beta_1 \frac{SI}{N} - dS$, $f_2 = \beta_1 \frac{SI}{N} - (a + d)E$, $f_3 = aE - (g_1 + d)I$, and $f_4 = g_1 I - dR$. Then, the Jacobian matrix is

$$J_{\text{SEIR}}(S, E, I, R) = \begin{pmatrix} -\beta_1 I/N - d & 0 & -\beta_1 S/N & 0 \\ \beta_1 I/N & -(a + d) & \beta_1 S/N & 0 \\ 0 & a & -(g_1 + d) & 0 \\ 0 & 0 & g_1 & -d \end{pmatrix}.$$

Evaluating at the endemic equilibrium, where $\beta_1 \frac{S^*}{N} = \frac{(a+d)(g_1+d)}{a}$, gives

$$\boxed{J^*_{\text{SEIR}} = \begin{pmatrix} -\beta_1 I^*/N - d & 0 & -\dfrac{(a+d)(g_1+d)}{a} & 0 \\ \beta_1 I^*/N & -(a+d) & \dfrac{(a+d)(g_1+d)}{a} & 0 \\ 0 & a & -(g_1+d) & 0 \\ 0 & 0 & g_1 & -d \end{pmatrix}}.$$

(c) **SIR model.** At a disease-free equilibrium, $S \approx N$, so the infected equation becomes $\dot{I} \approx \beta_2 I - (g_2 + d)I$. Hence, $\dot{I} = (\beta_2 - g_2 - d)I$. Therefore, the early exponential growth rate is

$$\boxed{r_{\text{SIR}} = \beta_2 - (g_2 + d)}.$$

SEIR model. At a disease-free equilibrium, $S \approx N$, so the exposed and infected equations become $\dot{E} \approx \beta_1 I - (a+d)E$ and $\dot{I} \approx aE - (g_1 + d)I$. Writing this system in matrix form gives

$$\begin{pmatrix} \dot{E} \\ \dot{I} \end{pmatrix} = \begin{pmatrix} -(a+d) & \beta_1 \\ a & -(g_1+d) \end{pmatrix} \begin{pmatrix} E \\ I \end{pmatrix}.$$

The early exponential growth rate is given by the dominant eigenvalue of this matrix. Hence, we solve

$$\det \begin{pmatrix} -(a+d) - r & \beta_1 \\ a & -(g_1+d) - r \end{pmatrix} = 0.$$

Therefore,

$$(r + a + d)(r + g_1 + d) - a\beta_1 = 0.$$

Expanding gives

$$r^2 + (a + g_1 + 2d)r + (a+d)(g_1+d) - a\beta_1 = 0.$$

Thus, solving for r via the quadratic equation,

$$r = \frac{-(a + g_1 + 2d) \pm \sqrt{(a + g_1 + 2d)^2 - 4\left[(a+d)(g_1+d) - a\beta_1\right]}}{2}.$$

Taking the larger root, the early exponential growth rate is

$$\boxed{r_{\text{SEIR}} = \frac{-(a + g_1 + 2d) + \sqrt{(a + g_1 + 2d)^2 - 4\left[(a+d)(g_1+d) - a\beta_1\right]}}{2}}.$$

(d) Suppose that the two models are parameterised such that they have the same basic reproduction number R_0.

(i) From part (b), for the SIR model, $S^* = \frac{N(g_2+d)}{\beta_2}$. Since $R_0^{\text{SIR}} = \frac{\beta_2}{g_2+d}$, this may be written as $S^* = \frac{N}{R_0}$. Similarly, for the SEIR model, $S^* = \frac{N(a+d)(g_1+d)}{a\beta_1}$. Since $R_0^{\text{SEIR}} = \frac{a\beta_1}{(a+d)(g_1+d)}$, we again have $S^* = \frac{N}{R_0}$. Therefore, if the two models have the same basic reproduction number R_0, then the equilibrium level of susceptible individuals is the same in both models.

(ii) In the SIR model, the number of new infections per unit time at equilibrium is $\beta_2 \frac{S^* I^*}{N}$. From the susceptible equation at equilibrium, $0 = dN - \beta_2 \frac{S^* I^*}{N} - dS^*$, so $\beta_2 \frac{S^* I^*}{N} = d(N - S^*)$. In the SEIR model, the number of new infections per unit time at equilibrium is $\beta_1 \frac{S^* I^*}{N}$. Again, from the susceptible equation at equilibrium, $0 = dN - \beta_1 \frac{S^* I^*}{N} - dS^*$, so $\beta_1 \frac{S^* I^*}{N} = d(N - S^*)$. Since part (i) showed that S^* is the same in both models when R_0 is the same, it follows that the number of new infections per unit time at equilibrium is also the same in both models.

(iii) Now assume that d is negligible compared to all other rates. In the SIR model, the average time from infection to recovery is approximately $\frac{1}{g_2}$. In the SEIR model, after infection, one spends an amount of time in E, then in I, so the average time from infection to recovery is approximately $\frac{1}{a} + \frac{1}{g_1}$. Let g_2 be chosen so that $\frac{1}{g_2} = \frac{1}{a} + \frac{1}{g_1} \implies g_2 = \frac{ag_1}{a+g_1}$, then, with equal R_0 values, $\beta_2 = g_2 R_0$ and $\beta_1 = g_1 R_0$. We, thus, see that $r_{\text{SIR}} = g_2(R_0 - 1) = \frac{ag_1}{a+g_1}(R_0 - 1)$. For the SEIR model, the positive root satisfies $f(r) := (r+a)(r+g_1) - ag_1 R_0 = 0$. Ultimately, $f(0) = ag_1(R_0 - 1) < 0$ for $R_0 > 1$. Substituting $r = r_{\text{SIR}}$ gives

$$f(r_{\text{SIR}}) = \frac{a^2 g_1^2 (R_0 - 1)^2}{(a + g_1)^2} > 0,$$

so the positive root must satisfy $0 < r_{\text{SEIR}} < r_{\text{SIR}}$.

2. (a) Analogous to the method used in part (a) of Question 1,

$$\boxed{R_0 = \frac{a\beta}{(a+d)(g+d)}}.$$

The mortality probability m does not affect R_0 because it acts only after individuals leave the infectious class, through the term $g(1-m)I$ in the recovered equation. Thus, m changes what happens after infection, but it does not change either the probability of reaching infectiousness or the expected time spent in the infectious class. Therefore, the expected number of secondary infections is unaffected by m.

(b) If a proportion p of newborn individuals is vaccinated immediately, then only a proportion $1 - p$ enters the susceptible class. Since the vaccine is perfectly (and immediately) effective, vaccinated individuals do not enter S. If there is a probability q of dying due to the vaccine, then a proportion $p(1-q)$ enters the recovered class and a proportion pq is lost due to vaccine mortality. The new model is therefore

$$\dot{S} = d(1-p) - \beta SI - dS,$$

$$\begin{aligned}\dot{E} &= \beta SI - aE - dE,\\ \dot{I} &= aE - gI - dI,\\ \dot{R} &= g(1-m)I + dp(1-q) - dR.\end{aligned}$$

To find the critical level of vaccination, we consider the disease-free equilibrium. Setting $E = I = 0$, the susceptible equation gives $S^* = 1 - p$. Hence, near the disease-free equilibrium, the exposed and infected subsystem is

$$\dot{E} = \beta(1-p)I - (a+d)E, \qquad \dot{I} = aE - (g+d)I.$$

Thus, the vaccinated reproduction number is $R_v = (1-p)R_0$. For eradication, we require $R_v < 1$, so $(1-p)R_0 < 1$. Therefore, the critical vaccination level is

$$\boxed{p_c = 1 - \frac{1}{R_0}}.$$

Hence, smallpox is eradicated provided that $p > p_c$.

(c) At the endemic steady state, we set $\dot{S} = \dot{E} = \dot{I} = \dot{R} = 0$. From the infected equation, $0 = aE^* - (g+d)I^*$, so

$$E^* = \frac{g+d}{a}I^*.$$

From the exposed equation, $0 = \beta S^* I^* - (a+d)E^*$. Substituting for E^* gives

$$\beta S^* I^* = (a+d)\frac{g+d}{a}I^*.$$

Since $I^* > 0$ at the endemic steady state, dividing by I^* gives

$$\beta S^* = \frac{(a+d)(g+d)}{a}.$$

Using $R_0 = \frac{a\beta}{(a+d)(g+d)}$, this becomes $S^* = \frac{1}{R_0}$. Now, from the susceptible equation, $0 = d(1-p) - \beta S^* I^* - dS^*$. Hence, $\beta S^* I^* = d(1 - p - S^*)$. Substituting $S^* = \frac{1}{R_0}$ gives

$$\frac{\beta}{R_0}I^* = d\left(1 - p - \frac{1}{R_0}\right),$$

so

$$I^* = \frac{dR_0}{\beta}\left(1 - p - \frac{1}{R_0}\right).$$

Therefore,

$$\boxed{I^* = \frac{d\left[(1-p)R_0 - 1\right]}{\beta}}.$$

The number of deaths due to vaccination per unit time is dpq, since a proportion p of newborn individuals is vaccinated and a proportion q of these die. The number of deaths due to infection per unit time is mgI^*, since infected individuals leave the infectious class at rate g and a proportion m die. We require deaths due to vaccination to be lower than deaths due to infection, so $dpq < mgI^*$. Substituting the steady-state value of I^* gives

$$dpq < mg \cdot \frac{d\left[(1-p)R_0 - 1\right]}{\beta}.$$

Cancelling d from both sides yields

$$pq < \frac{mg\left[(1-p)R_0 - 1\right]}{\beta},$$

and, ultimately,

$$\boxed{q < mg\,\frac{(1-p)R_0 - 1}{p\beta}}.$$

Thus, the number of deaths due to vaccination is lower than the number of deaths due to infection whenever $q < mg\,\frac{(1-p)R_0-1}{p\beta}$.

(d) If vaccination is maintained at a level that keeps smallpox eradicated, then there are no infected individuals at steady state, so there are no deaths due to smallpox infection. Hence, the only deaths are those due to vaccination. From part (b), the critical vaccination level is

$$p_c = 1 - \frac{1}{R_0}.$$

To maintain eradication with the fewest possible vaccine-related deaths, we choose the minimum vaccination level required for eradication, namely $p = p_c$. Since newborn individuals enter at rate d, the number vaccinated per unit time is dp_c and the number of deaths due to vaccination per unit time is, therefore, dp_cq. Therefore, over a time period Δt, the minimum number of deaths is

$$\boxed{D_{\min} = dp_cq\,\Delta t = dq\left(1 - \frac{1}{R_0}\right)\Delta t}.$$

This provides a rationale for recommending vaccination as a public health policy even when $q > 0$. Although the vaccine carries a positive risk of death, maintaining vaccination at the critical level prevents deaths due to smallpox altogether. Since smallpox is a severe infectious disease, the mortality burden avoided by eradication can outweigh the comparatively small mortality risk associated with vaccination. Moreover, choosing $p = p_c$ minimises vaccine-related deaths while still ensuring that transmission cannot be sustained.

3. (a) If a proportion p of all newborn individuals is vaccinated, then only a proportion $1 - p$ enters the susceptible class, while a proportion p enters the recovered class immediately. The model therefore becomes

$$\dot{S} = d(1 - p) - \beta SI - dS,$$

$$\dot{I} = \beta SI - gI - dI,$$

$$\dot{R} = dp + gI - dR.$$

(b) Firstly. consider the equilibrium level of infection. At an endemic equilibrium $I^* > 0$, so from $\dot{I} = 0$ we have $\beta S^* - (g+d) = 0$, hence $S^* = \frac{g+d}{\beta}$. Using $\dot{S} = 0$, we obtain $d(1-p) - \beta S^* I^* - dS^* = 0$, so $\beta S^* I^* = d(1-p-S^*)$. Substituting for S^* gives

$$I^* = \frac{d}{\beta S^*}(1 - p - S^*) = \frac{d}{g+d}\left(1 - p - \frac{g+d}{\beta}\right)$$

Therefore, $I^* = 0$ when $1 - p - \frac{g+d}{\beta} = 0$, that is when $p = 1 - \frac{g+d}{\beta}$. Now, consider the growth rate of infection at the disease-free equilibrium. At the disease-free equilibrium $I \approx 0$ and $S^* = 1-p$, so the infected equation becomes

$$\dot{I} \approx (\beta(1 - p) - (g + d))I$$

Thus, the early growth rate is $r = \beta(1 - p) - (g + d)$. This reaches zero when $\beta(1 - p) - (g + d) = 0$, so again $p = 1 - \frac{g+d}{\beta}$. Therefore, both the equilibrium level of infection and the growth rate of infection reach zero at the same critical vaccination level

$$\boxed{p_c = 1 - \frac{g+d}{\beta}}.$$

Since $R_0 = \frac{\beta}{g+d}$ for this model, this may also be written as $\boxed{p_c = 1 - \frac{1}{R_0}}$.

(c) Let $H(p)$ denote the total hospitalisation rate in the population. Vaccination causes hospitalisation with probability h_V, so the hospitalisation rate due to vaccination is dph_V. For infection, we use the equilibrium infection rate. From part (b), when $p < p_c$ the endemic equilibrium satisfies $S^* = \frac{g+d}{\beta} = \frac{1}{R_0}$ and $\beta S^* I^* = d(1 - p - S^*)$, so the rate of new infections is $d\left(1-p-\frac{1}{R_0}\right)$. The hospitalisation rate due to infection is, hence, $dh_I\left(1-p-\frac{1}{R_0}\right)$. Therefore, for $p < p_c$,

$$H(p) = dp\,h_V + dh_I\left(1-p-\frac{1}{R_0}\right) = d\left[h_I\left(1-\frac{1}{R_0}\right) + p(h_V - h_I)\right].$$

Since $h_V < h_I \implies h_V - h_I < 0$, $H(p)$ is decreasing in p for $p < p_c$. Thus, on this interval, the minimum is attained at the largest possible value of p, namely $p = p_c$. For $p \geq p_c$, infection is eradicated, so there are no infection-related hospitalisations and $H(p) = dph_V$, which is increasing in p. Hence, on this interval, the minimum is attained at the smallest possible value of p, again, $p = p_c$. Ultimately, the optimum level of vaccination that minimises the hospitalisation rate in the entire population is

$$\boxed{p^* = p_c = 1 - \frac{g+d}{\beta} = 1 - \frac{1}{R_0}}.$$

Thus, the socially optimal strategy is to vaccinate exactly at the critical threshold; any lower level leaves unnecessary hospitalisations due to infection, while any higher level creates unnecessary hospitalisations due to vaccination.

(d) Suppose the rest of the population vaccinates at level p. Following the control framework in the lecture notes, we treat a single individual as negligible, so their own vaccination choice does not change the population-level prevalence determined by p. Thus, the individual compares the hospitalisation risk from vaccination with the hospitalisation risk from remaining unvaccinated in a population where the endemic level of infection is set by p. If the individual vaccinates, the probability of hospitalisation is simply h_V. If the individual does not vaccinate, then, while susceptible, they are infected at rate βI^* and die naturally at rate d. Hence, the probability that an unvaccinated individual is infected before natural death is

$$\frac{\beta I^*}{\beta I^* + d},$$

so the probability of hospitalisation without vaccination is

$$h_I\frac{\beta I^*}{\beta I^* + d}.$$

Therefore, if an individual vaccinates with probability q, their total risk of hospitalisation is

$$H(q;p) = qh_V + (1-q)h_I\frac{\beta I^*}{\beta I^* + d}.$$

From part (c), for $p < p_c$ we have $I^* = \frac{d((1-p)R_0-1)}{\beta}$, so $\beta I^* = d\left[(1-p)R_0 - 1\right]$. Substituting this gives

$$\begin{aligned} H(q;p) &= qh_V + (1-q)h_I \frac{(1-p)R_0 - 1}{(1-p)R_0} \\ &= qh_V + (1-q)h_I \left(1 - \frac{1}{(1-p)R_0}\right). \end{aligned}$$

This is linear in q, so the minimum occurs at one of the endpoints:

- $q^* = 1$ if $h_V < h_I \left(1 - \frac{1}{(1-p)R_0}\right)$,
- $q^* = 0$ if $h_V > h_I \left(1 - \frac{1}{(1-p)R_0}\right)$,
- any $q \in [0,1]$ if $h_V = h_I \left(1 - \frac{1}{(1-p)R_0}\right)$.

If $p \geq p_c = 1 - \frac{1}{R_0}$, then the disease is eradicated and $I^* = 0$, so $H(q;p) = qh_V$. In this case, the individual risk is minimised by choosing $\boxed{q^* = 0}$. Therefore, the probability of vaccination that minimises an individual's risk of hospitalisation is

$$\boxed{q^* = \begin{cases} 1, & \text{if } h_V < h_I \left(1 - \dfrac{1}{(1-p)R_0}\right), \\ 0, & \text{if } h_V > h_I \left(1 - \dfrac{1}{(1-p)R_0}\right), \end{cases}}$$

with indifference when $h_V = h_I \left(1 - \frac{1}{(1-p)R_0}\right)$.

(e) A vaccination level p is an evolutionary stable strategy (ESS) if the risk of hospitalisation is the same for all q, so the individual is indifferent between vaccinating and not vaccinating. For this to be the same for all q, the coefficient of q must be zero. Equivalently, the hospitalisation risk from vaccinating must equal the hospitalisation risk from not vaccinating, so $h_V = h_I \left(1 - \frac{1}{(1-p)R_0}\right)$. Rearranging gives

$$\frac{h_V}{h_I} = 1 - \frac{1}{(1-p)R_0},$$

so

$$\frac{1}{(1-p)R_0} = 1 - \frac{h_V}{h_I}.$$

Hence,

$$(1-p)R_0 = \frac{1}{1 - \frac{h_V}{h_I}}$$

and, ultimately,

$$1 - p = \frac{1}{R_0\left(1 - \frac{h_V}{h_I}\right)},$$

where $R_0\left(1 - \frac{h_V}{h_I}\right) > 0$. Thus, the ESS is

$$\boxed{p_{\text{ESS}} = 1 - \frac{1}{R_0\left(1 - \frac{h_V}{h_I}\right)}}.$$

From part (c), the optimum level of vaccination for the whole population is $p^* = 1 - \frac{1}{R_0}$. Since $0 < h_V < h_I$, we have $0 < 1 - \frac{h_V}{h_I} < 1$, so $R_0\left(1 - \frac{h_V}{h_I}\right) < R_0$. Therefore,

$$\frac{1}{R_0\left(1 - \frac{h_V}{h_I}\right)} > \frac{1}{R_0}$$

and $p_{\text{ESS}} < p^*$. Therefore, the ESS is lower than the socially optimal vaccination level. The interpretation is that the level in part (c) is the *population optimum*: it minimises the total hospitalisation burden across the whole population. By contrast, the level in part (d) describes the *best response* of a single individual when the rest of the population vaccinates at level p. The ESS found here is the population vaccination level at which individuals are exactly indifferent between vaccinating and not vaccinating, so nobody has an incentive to change their behaviour. Since individuals do not fully account for the protection their vaccination provides to others, the ESS is lower than the population optimum.

(f) Following the lecture notes, for prophylactic vaccination with imperfect protection, we can interpret p as coverage multiplied by efficacy. Hence, if only a fraction π of vaccinations is successful, the effective protected fraction is $p\pi$, rather than p.

(i) If a proportion π of the vaccinated population is perfectly protected and the remaining vaccinated individuals are not protected at all, then among newborns a proportion $p\pi$ enters R immediately, while the remaining proportion $1 - p\pi$ enters S. Thus, the model becomes $\dot{S} = d(1-p\pi) - \beta SI - dS$, $\dot{I} = \beta SI - gI - dI$, and $\dot{R} = dp\pi + gI - dR$.

(ii) If each vaccinated individual has probability π of being protected, then at the population level the expected fraction successfully immunised is again $p\pi$. Hence the deterministic model is unchanged: $\dot{S} = d(1-p\pi) - \beta SI - dS$, $\dot{I} = \beta SI - gI - dI$ and $\dot{R} = dp\pi + gI - dR$. In part (i), exactly a proportion π of vaccinated individuals is protected, whereas in part (ii) protection is assigned independently to each vaccinated individual.

4. (i) At an early stage of the infection, we assume $S(t) \approx 1$, so the model becomes $\dot{E} = \beta I - \sigma E$ and $\dot{I} = \sigma E - \gamma I$. Since the observed quantity is $y(t) = \rho\sigma E(t)$, we have $\dot{y}(t) = \rho\sigma\dot{E}(t) = \rho\sigma(\beta I - \sigma E) = \rho\beta\sigma I - \sigma y$. Hence, $\rho\beta\sigma I = \dot{y} + \sigma y$. Differentiating again gives $\ddot{y}(t) = \rho\beta\sigma\dot{I}(t) - \sigma\dot{y}(t)$. Using $\dot{I} = \sigma E - \gamma I$, this becomes $\ddot{y} = \rho\beta\sigma(\sigma E - \gamma I) - \sigma\dot{y}$. Since $y = \rho\sigma E$, we have $\rho\beta\sigma E = \beta\sigma y$; furthermore, since $\rho\beta\sigma I = \dot{y} + \sigma y$, we obtain $\ddot{y} = \beta\sigma y - \gamma(\dot{y} + \sigma y) - \sigma\dot{y}$. Therefore, $\ddot{y} = -(\sigma + \gamma)\dot{y} + \sigma(\beta - \gamma)y$, so y satisfies the second-order differential equation

$$\boxed{\ddot{y} + (\sigma + \gamma)\dot{y} - \sigma(\beta - \gamma)y = 0}.$$

To solve this, let $y(t) = e^{\lambda t}$. Then, the characteristic equation is $\lambda^2 + (\sigma + \gamma)\lambda - \sigma(\beta - \gamma) = 0$, with roots

$$\lambda_{1,2} = \frac{-(\sigma + \gamma) \pm \sqrt{(\sigma - \gamma)^2 + 4\sigma\beta}}{2},$$

via the quadratic equation. Due to the discriminant being strictly positive for these parameters, the general solution is $y(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$. Using $y(0) = 0$, we see that $C_1 + C_2 = 0$, so $C_2 = -C_1$ and $y(t) = C_1\left(e^{\lambda_1 t} - e^{\lambda_2 t}\right)$. Now, $\dot{y}(t) = C_1\left(\lambda_1 e^{\lambda_1 t} - \lambda_2 e^{\lambda_2 t}\right)$, so, from $\dot{y}(0) = 1$, we obtain $C_1(\lambda_1 - \lambda_2) = 1 \implies C_1 = \frac{1}{\lambda_1 - \lambda_2}$. It follows that

$$\boxed{y(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}},$$

where $\lambda_{1,2}$ is defined above.

(ii) From observations of y, we can identify only the two roots λ_1 and λ_2 from the quantities $\sigma + \gamma$ and $\sigma(\beta - \gamma)$ via Vieta's formula. Therefore, the three parameters β, σ and γ are *not* separately identifiable from y alone, since different triples (β, σ, γ) can give the same values of $\sigma + \gamma$ and $\sigma(\beta - \gamma)$ and, ultimately, the same function $y(t)$. Moreover, ρ is also not separately identifiable from y, since y only observes the product of $\rho\sigma E$. Thus, the identifiable quantities from y are only $\boxed{\lambda_1 \text{ and } \lambda_2}$.

(iii) See `CW1_4iii_4iv.Rmd` for the code used and plots generated. Assume $S(t) \approx 1$. In part (i), we introduced artificial normalisation through the initial conditions. However, as we are now modelling real data, we introduce a scale parameter C such that $y(t) = C\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}$. As the dataset contains noisy observations of $(y(t))^2$, the fitted mean curve is of the form $\hat{z}_i \approx \mu(t_i; C, \lambda_1, \lambda_2) := \left(C\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2}\right)^2$. Thus, we opt for nonlinear least squares analysis. The parameters C, λ_1 and λ_2 are estimated by minimising the residual sum of squares (RSS): $\mathrm{RSS}(C, \lambda_1, \lambda_2) = \sum_{i=1}^{n} \left(z_i - \mu(t_i; C, \lambda_1, \lambda_2)\right)^2$. As this approach is sensitive to starting values, we find an initial guess of the dominant growth rate. For sufficiently large t in the early growth phase, the larger root usually dominates, so $y(t) \approx Ke^{\lambda_1 t} \implies (y(t))^2 \approx K^2 e^{2\lambda_1 t}$. Hence, $\log(z(t))$ should be approximately linear in time, with slope $2\lambda_1$. We, thus, fit a rudimentary linear model to construct a starting value for λ_1. For the second root, a small negative value seems appropriate (as the second mode is often decaying or much less dominant early in the epidemic). The fitted dominant root $\hat{\lambda}_1 \approx 0.2139$ represents the early growth of the epidemic. As $\hat{\lambda}_1 > 0$, the model predicts initial exponential expansion of the outbreak, with an approximate doubling time of $T_d = \frac{\ln(2)}{\hat{\lambda}_1} \approx 3.2$ days, indicating relatively rapid early spread. The second root, $\hat{\lambda}_2 \approx -0.3280$, corresponds to a decaying transient mode; its effects are strongest at very early times and dies out quickly. The identifiable combinations $\widehat{\sigma + \gamma} \approx 0.1141$ and $\widehat{\sigma(\beta - \gamma)} \approx 0.0701$ summarise the combined effects of progression through the exposed and infectious stages and the net transmission pressure in the early growth phase. In particular, the positive estimate of $\sigma(\beta-\gamma)$ is consistent with transmission exceeding recovery strongly enough to generate epidemic growth. Since the dominant growth rate is positive, the fitted early-stage dynamics are consistent with a *supercritical* outbreak (although R_0 itself is not identifiable from the data alone). The residual plot shows clear fanning (heteroskedasticity), which is expected due to nonlinearity; nevertheless, the residuals remain broadly centered around zero with no systematic curvature, suggesting that the fitted mean curve captures the main early epidemic trend reasonably well.

(iv) See `CW1_4iii_4iv.Rmd` for the code used and plots generated. Again, assume $S(t) \approx 1$. We fit an exponential model of the form $g(t) := a + be^{ct}$, as well as three n-th order polynomials, $n = 2, 3, 4$. The exponential model produced a slightly smaller RSS value than the original (mechanistic) model (20461.44 and 20501.58, respectively). The quadratic model performed notably worse, with a larger RSS value and slight curvature in the residual plot. This suggests that the observed data exhibit approximately exponential growth during the early epidemic phase, consistent with the dominant growth mode identified in the previous part. The cubic and quartic fits achieved the lowest RSS, although the improvement from cubic to quartic was negligible, indicating the the quartic model provides very little additional explanatory power and may, simply, be fitting noise. We conclude that a polynomial of order 3 is the most suitable. Overall, the similarity between the original and exponential models suggests that the inferred rapid early growth is reasonably robust to model choice. In particular, the positive, dominant growth rate and implied short doubling time appear to be stable, qualitative features of the data. However, the precise parameter values obtained in the previous part should be interpreted cautiously when informing public-health policy; these estimates depend on the early-stage approximation $S(t) \approx 1$, are derived from a limited window of noisy observations and rely on a specific mechanistic model structure. Moreover, the polynomial fits demonstrate that several functional forms can reproduce the observed trajectory over the same interval very closely (as demonstrated in the fitted plot), telling us that the exact numerical parameter estimates are not highly robust.